

Part 1: Theoretical Questions

Q1.1

1. Find MGUs for the following pairs of type expressions, if such an MGU exists [3 points each]:

(a) $[T1 * [T1 \rightarrow T2] \rightarrow N]$, $[[T3 \rightarrow T4] * [T5 \rightarrow \text{Number}] \rightarrow N]$

MGU: $\{T1 = [T3 \rightarrow T4], T5 = [T3 \rightarrow T4], T2 = \text{Number}\}$

(b) $[T1 * [T1 \rightarrow T2] \rightarrow N]$, $[\text{Number} * [\text{Symbol} \rightarrow T4] \rightarrow N]$

MGU: There are no unifier (because we can get $T1 = \text{Number}$ and $T1 = \text{Symbol}$)

(c) $T1, T2$

MGU: $\{T1 = T2\}$

(d) $\text{Boolean}, \text{Boolean}$

MGU: $\{\}$

2. Are the following typing statements true or false? Explain why (you might want to consult the guide published under “typing statements summary”) [3 points each]:

(a) $\{f: [T2 \rightarrow T3], g: [T1 \rightarrow T2], a: \text{Boolean}\} \vdash (f (g a)): T3$

Answer: true, $T1 = \text{Boolean}$, and the resulting statement is correct.

(b) $\{f: [T2 \rightarrow T1], x: T1, y: T3\} \vdash (f x): T1$

Answer: false, f has type $T2 \rightarrow T1$ while x has type $T1$ so $(f x)$ expects an argument of type $T2$. So the application $(f x)$ is not type correct.

Q1.2

In the type inference implementation, we represent type variables (TVar) with a content field (which is a Box which contains a type expression value, or #f when empty). In this representation, we can have a TVar refer in its content to another TVar iteratively, leading to a chain of TVars. Write a program which, when we pass it to the type inference algorithm, creates a chain of length five, i.e., $Tvar1 \rightarrow Tvar2 \rightarrow Tvar3 \rightarrow Tvar4 \rightarrow Tvar5$. [7 points]

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(define f1 (lambda (a) a))
(define f2 (lambda (b) (f1 b)))
(define f3 (lambda (c) (f2 c)))
(define f4 (lambda (d) (f3 d)))
(define f5 (lambda (e) (f4 e)))
(f5 2)
```